Fourier Transforms and Discrete-Time Fourier Transforms for Periodic Signals

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Key Concepts

1) Neither the Fourier transform (FT) nor the discrete-time Fourier transform (DTFT) exist in a strict sense for periodic signals because the integral definition for $X(\Omega)$ and the infinite series definition for $X(e^{j\omega})$ do not converge.

2) A FT or DTFT for periodic signals is defined by using impulses in a way that satisfies all the properties of the FT and DTFT.

3) The primary purpose for defining a FT and DTFT for periodic signals is to perform frequency-domain analysis of interactions between periodic and non periodic signals.

4) FT pairs for periodic signals

a)

$$e^{j\Omega_o t} \overset{FT}{\longleftrightarrow} 2\pi\delta(\Omega - \Omega_o)$$

b)

$$\cos(\Omega_o t) \overset{FT}{\longleftrightarrow} \pi\delta(\Omega - \Omega_o) + \pi\delta(\Omega + \Omega_o)$$

c)

$$\sin(\Omega_o t) \overset{FT}{\longleftrightarrow} \frac{\pi}{j}\delta(\Omega - \Omega_o) - \frac{\pi}{j}\delta(\Omega + \Omega_o)$$

d)

$$\sum_{k=-\infty}^{\infty} X[k]e^{jk\Omega_o t} \xleftrightarrow{FT} 2\pi \sum_{k=-\infty}^{\infty} X[k]\delta(\Omega - k\Omega_o)$$

e)

$$\sum_{m=-\infty}^{\infty} \delta(t - mT) \xleftrightarrow{FT} 2\pi \sum_{k=-\infty}^{\infty} \frac{1}{T}\delta(\Omega - k\frac{2\pi}{T})$$

5) DTFT pairs for periodic signals specified for $-\pi \leq \omega < \pi$ and assuming $-\pi \leq \omega_o < \pi$

a)

$$e^{j\omega_o n} \overset{DTFT}{\longleftrightarrow} 2\pi\delta(\omega - \omega_o)$$

b)

$$\cos(\omega_o n) \overset{DTFT}{\longleftrightarrow} \pi\delta(\omega - \omega_o) + \pi\delta(\omega + \omega_o)$$

c)

$$\sin(\omega_o n) \overset{DTFT}{\longleftrightarrow} \frac{\pi}{j}\delta(\omega - \omega_o) - \frac{\pi}{j}\delta(\omega + \omega_o)$$

The FT and DTFT do not exist in a strict sense for periodic signals

Eg. if $x(t) = \sin(2\pi t)$, $X(\Omega) = \int_{-\infty}^{\infty} \sin(2\pi t) e^{j\Omega t} dt$ doesn't converge

We can define a FT and a DTFT for periodic signals by incorporating impulses

- the defining equations for the transform don't converge
- the transforms satisfy all the properties and thus are useful

The FT and DTFT for periodic signals are used to analyze interactions between periodic and non periodic signals

Approach: 1) Find transform for a complex sinusoid

2) Express periodic signal as a sum of complex sinusoids

3) Apply linearity

Suppose $X(\Omega) = 2\pi\delta(\Omega) \rightarrow x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\delta(\Omega)e^{j\Omega t}d\Omega = 1$

Similarly $Y(\Omega) = 2\pi\delta(\Omega-\Omega_0) \rightarrow y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\delta(\Omega-\Omega_0)e^{j\Omega t}d\Omega = e^{j\Omega_0 t}$

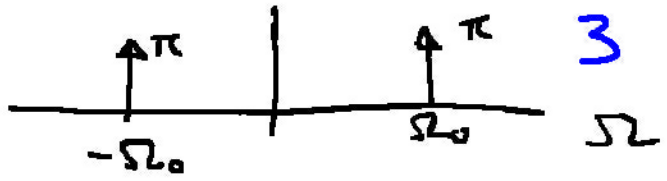
Frequency shift property relates $Y(\Omega)$ to $X(\Omega)$

$$e^{j\Omega_0 t} \overset{FT}{\longleftrightarrow} 2\pi\delta(\Omega-\Omega_0)$$

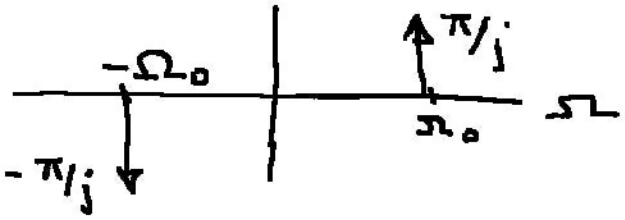
$$\cos(\Omega_0 t) = \frac{1}{2}e^{j\Omega_0 t} + \frac{1}{2}e^{-j\Omega_0 t} \overset{FT}{\longleftrightarrow} \pi\delta(\Omega-\Omega_0) + \pi\delta(\Omega+\Omega_0)$$

$$\sin(\Omega_0 t) = \frac{1}{2j}e^{j\Omega_0 t} - \frac{1}{2j}e^{-j\Omega_0 t} \overset{FT}{\longleftrightarrow} \frac{\pi}{j}\delta(\Omega-\Omega_0) - \frac{\pi}{j}\delta(\Omega+\Omega_0)$$

$\cos(\Omega_0 t) \overset{FT}{\longleftrightarrow} \pi\delta(\Omega-\Omega_0) + \pi\delta(\Omega+\Omega_0)$



$\sin(\Omega_0 t) \overset{FT}{\longleftrightarrow} \frac{\pi}{j}\delta(\Omega-\Omega_0) - \frac{\pi}{j}\delta(\Omega+\Omega_0)$

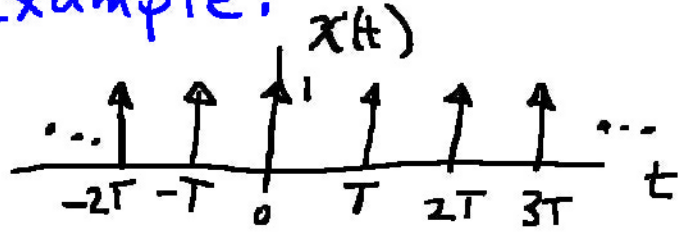


In general

$$x(t) = \sum_{k=-\infty}^{\infty} X[k]\, e^{jk\Omega_0 t} \overset{FT}{\longleftrightarrow} X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} X[k]\,\delta(\Omega - k\Omega_0)$$

(Fourier Series)

Example:



$$x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT)$$

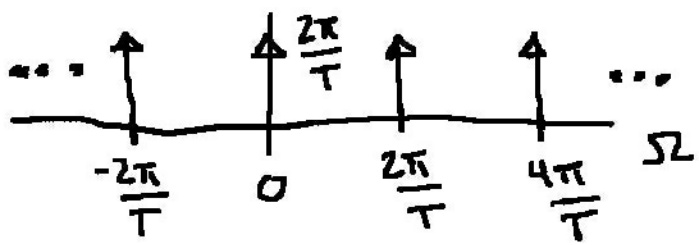
Find $x(t) \overset{FS;\Omega_0}{\longleftrightarrow} X[k]$

$\Omega_0 = \frac{2\pi}{T}$

$$X[k] = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)\, e^{-jk\Omega_0 t}\, dt$$

$$= \frac{1}{T}$$

$$X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} \frac{1}{T}\delta\left(\Omega - k\frac{2\pi}{T}\right)$$



Similarly $X(e^{j\omega}) = 2\pi\delta(\omega-\omega_0)$; $-\pi < \omega_0 < \pi$

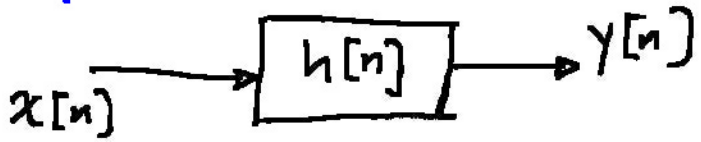
$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} 2\pi\,\delta(\omega-\omega_0)\,e^{j\omega n}\,d\omega = e^{j\omega_0 n}$$

$e^{j\omega_0 n} \xleftrightarrow{DTFT} 2\pi\delta(\omega-\omega_0)$

$\cos(\omega_0 n) \xleftrightarrow{DTFT} \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$

$\sin(\omega_0 n) \xleftrightarrow{DTFT} \frac{\pi}{j}\delta(\omega-\omega_0) - \frac{\pi}{j}\delta(\omega+\omega_0)$

Example:

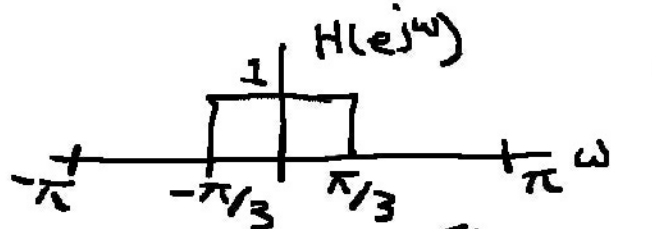


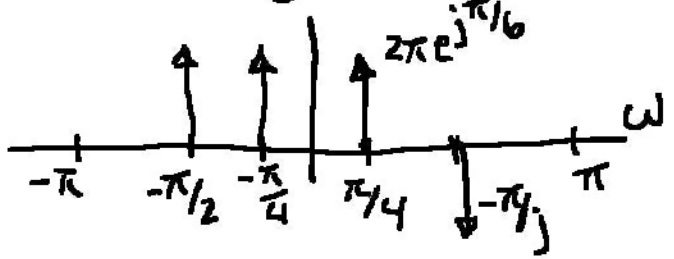
$$h[n] = \frac{\sin(\pi/3\, n)}{\pi n}$$

$$x[n] = 2\cos\left(\frac{\pi}{4}n + \frac{\pi}{6}\right) - \sin\left(\frac{\pi}{2}n\right)$$

Find $y[n]$.

$Y(e^{j\omega}) = H(e^{j\omega})\,X(e^{j\omega})$





$$X(e^{j\omega}) = 2\pi e^{j\frac{\pi}{6}}\delta(\omega-\tfrac{\pi}{4}) + 2\pi e^{-j\pi/6}\delta(\omega+\tfrac{\pi}{4}) - \frac{\pi}{j}\delta(\omega-\tfrac{\pi}{2}) + \frac{\pi}{j}\delta(\omega+\tfrac{\pi}{2})$$

$$y[n] = 2\cos\left(\frac{\pi}{4}n + \frac{\pi}{6}\right)$$

$$Y(e^{j\omega}) = 2\pi e^{j\pi/6}\delta(\omega-\tfrac{\pi}{4}) + 2\pi e^{-j\pi/6}\delta(\omega+\tfrac{\pi}{4})$$